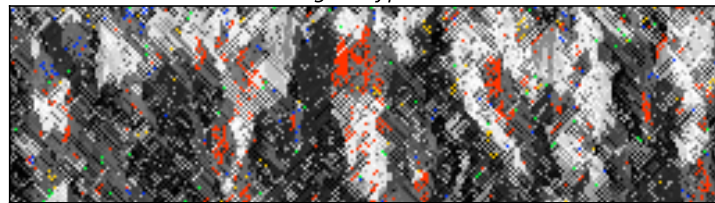
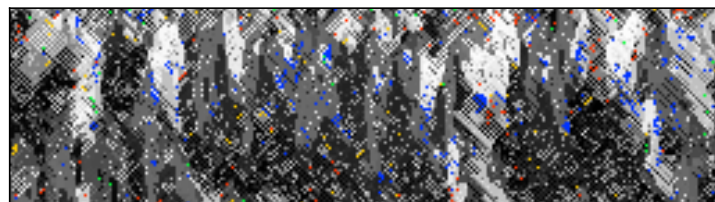


(a) Reaching high diversity: four mechanisms that share no machinery

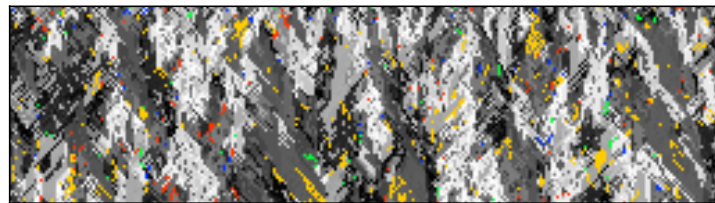
genotype



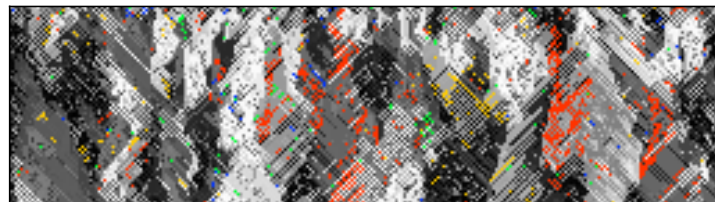
braid P_a / two-4-cycle — temporal alternation (causal reach 3.23 cells)



braid rot+2 / rot+4 — two operators that collapse alone (causal reach 3.81 cells)



P_a , blend strength 0.70 — continuous blending (causal reach 3.60 cells)



P_a / two-4-cycle niches, width 64 — spatial partition (causal reach 2.72 cells)

phenotype

